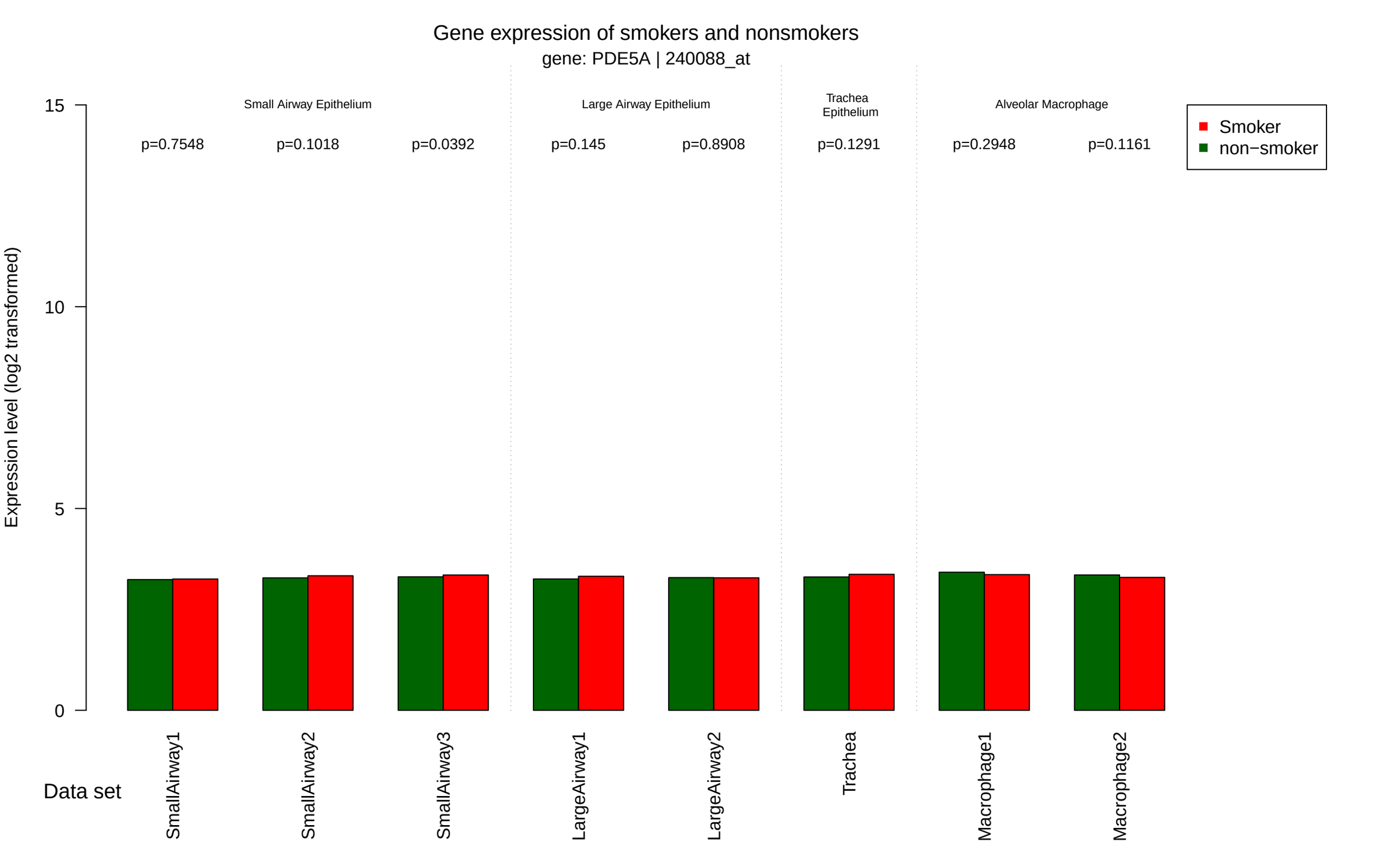
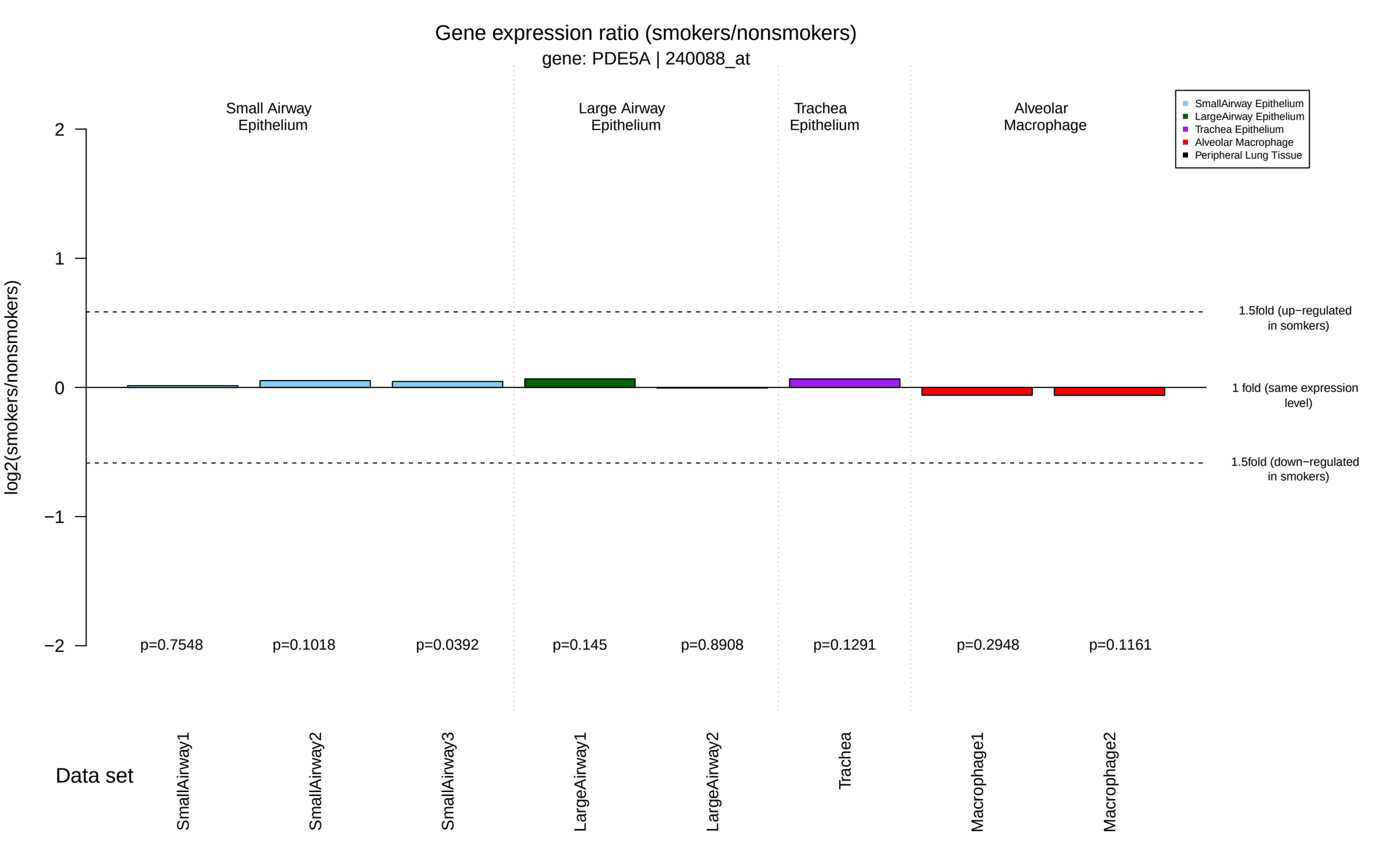
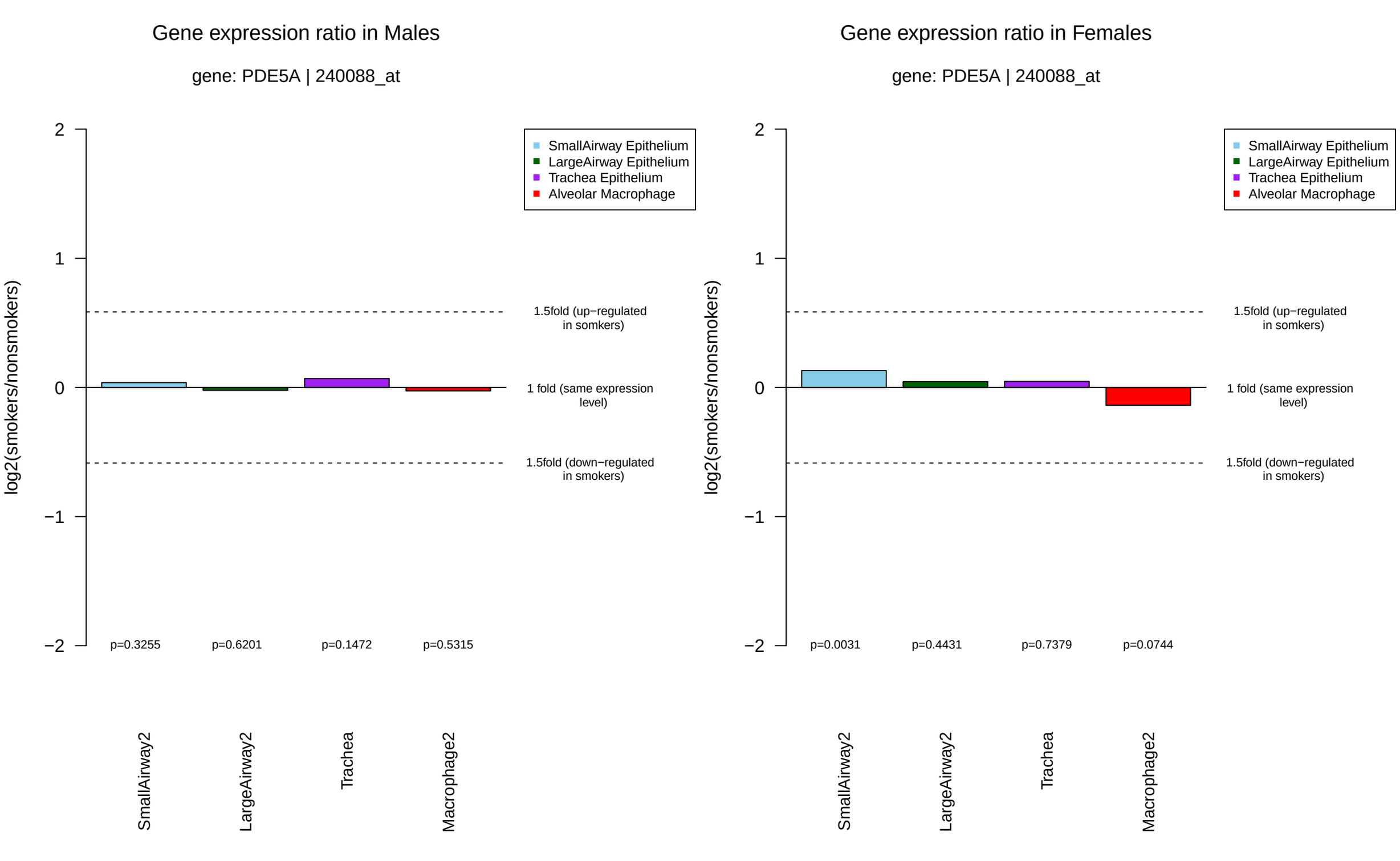
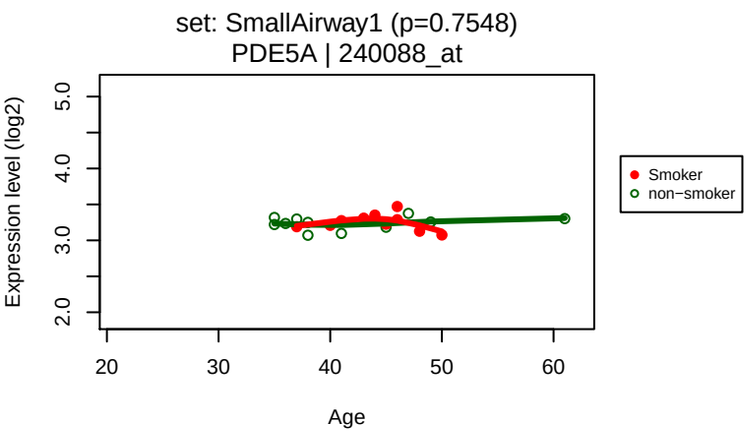
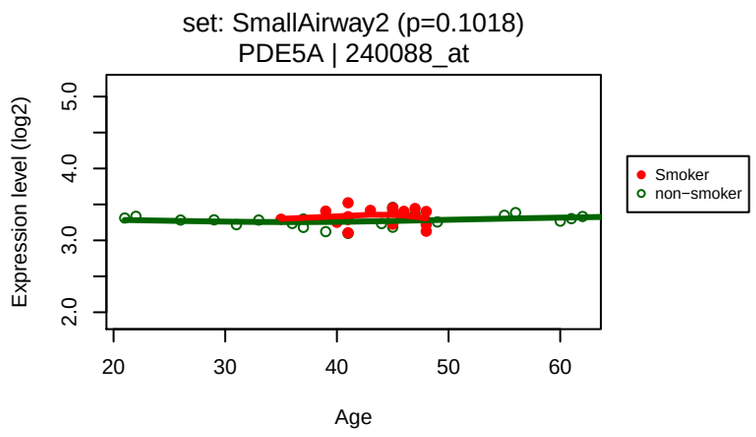
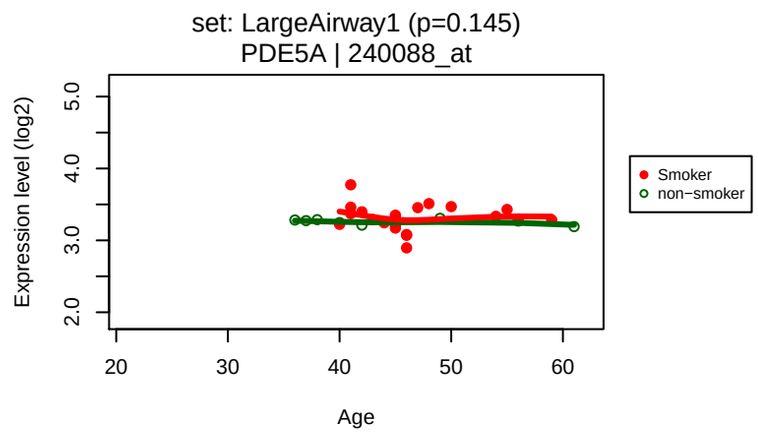

Gene expression of smokers and nonsmokers
gene: PDE5A | 240088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7548
p=0.1018
p=0.0392
p=0.145
p=0.8908
p=0.1291
p=0.2948
p=0.1161
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

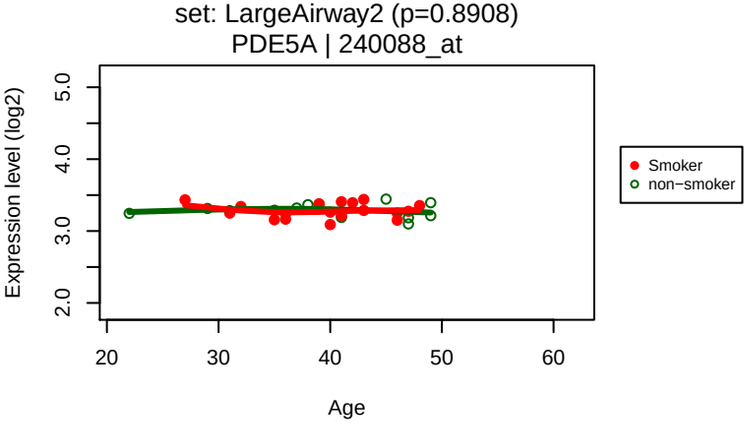

Gene expression ratio (smokers/nonsmokers)
gene: PDE5A | 240088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7548
p=0.1018
p=0.0392
p=0.145
p=0.8908
p=0.1291
p=0.2948
p=0.1161
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

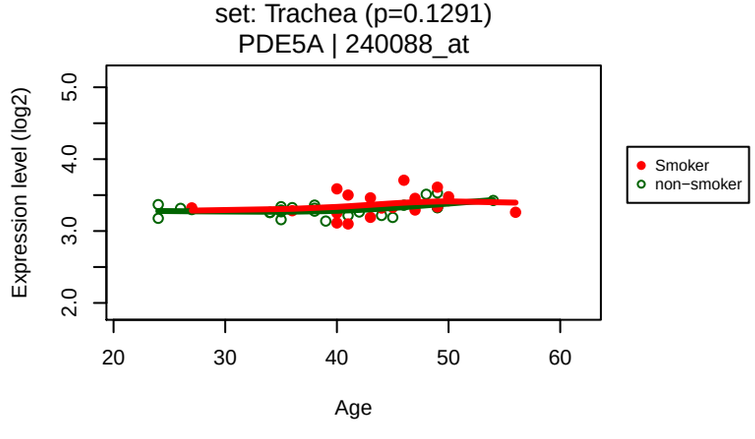

Gene expression ratio in Males
gene: PDE5A | 240088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3255
p=0.6201
p=0.1472
p=0.5315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE5A | 240088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.4431
p=0.7379
p=0.0744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

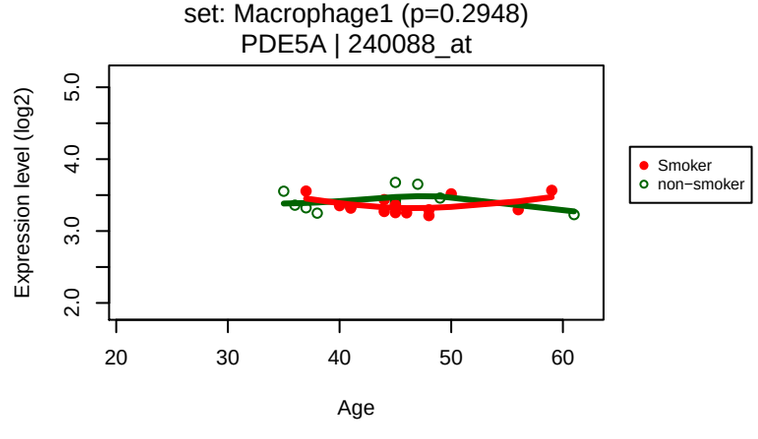

set: SmallAirway1 (p=0.7548)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker

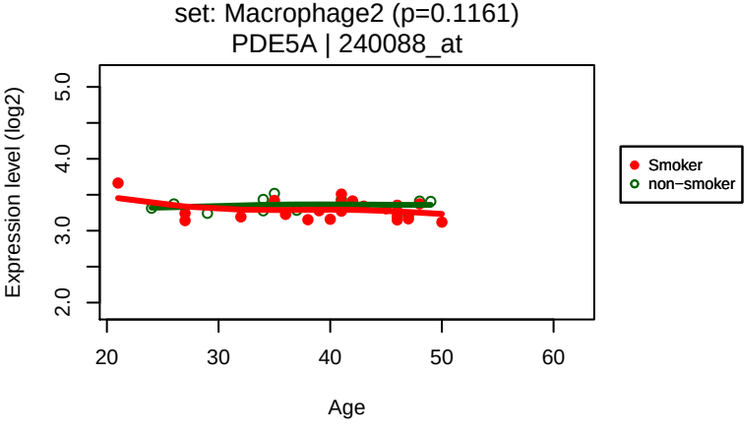

set: SmallAirway2 (p=0.1018)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker

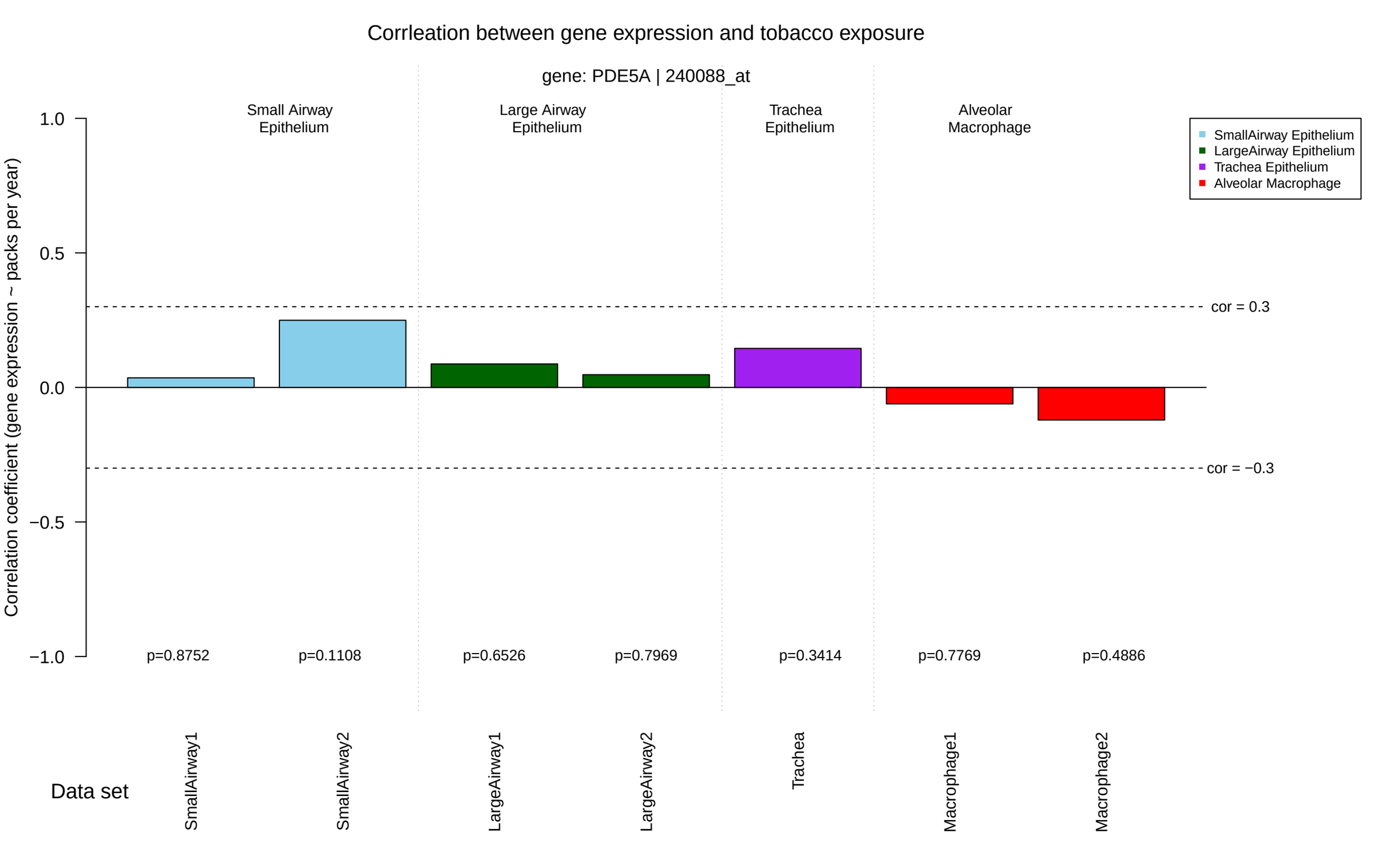

set: LargeAirway1 (p=0.145)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8908)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1291)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2948)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1161)
PDE5A | 240088_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE5A | 240088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8752
p=0.1108
p=0.6526
p=0.7969
p=0.3414
p=0.7769
p=0.4886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2